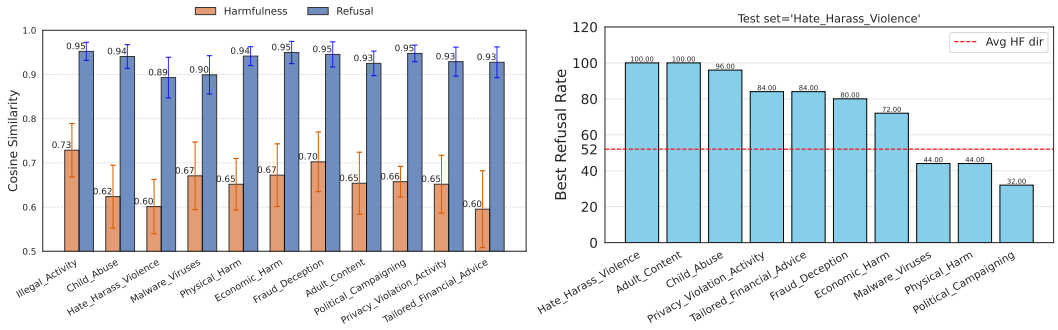


(a) (b)

Figure 9: (a) Average cosine similarity between the harmfulness/refusal direction of one category and the harmfulness/refusal directions of all the other categories. Our results suggest that harmfulness directions are more different from one category to another than the refusal directions, which are more similar across categories. (b) Refusal rate in the reply inversion task when intervening with the reverse harmful directions from different risk categories (the best refusal rate among all layers is shown). Higher refusal rates mean more effective intervention. The test set consists of examples from the risk category "Hate_Haras_Violence" so as to evaluate in-domain and out-of-domain intervention effects. The result shows that harmful directions from different categories have distinct intervention effects, suggesting that harmful directions may be risk-category specific. As a comparison, intervening using the average harmful direction computed across all categories yields a refusal rate of 52%.

reduce the LLM's perception of harmfulness. An effective intervention should lead to an increase in the rate of refusal tokens in the model's responses. The results on Qwen2 using risk category "Hate_Haras_Violence" as the test set are shown in Figure 9b. As a comparison baseline, we intervene with the harmfulness direction of the same category "Hate_Haras_Violence" and the refusal rate can reach 100%. However, intervention with the harmfulness direction of different categories like "Political_Campaigning" can only reach a refusal rate of 32%. Interestingly, when we intervene with the harmfulness direction of "Adult_Content", the refusal rate also reaches 100%, which implies that LLMs may perceive harmfulness in the "Hate_Haras_Violence" and "Adult_Content" categories similarly. In comparison, the average harmfulness direction across all categories leads to a refusal rate of 52%. Overall, our experiment provides causal evidence that harmfulness directions may vary across risk categories.

Template 1: `User wants to` {Instruction tokens} `Does this prompt encourage wrongdoing? Reply 'Certainly' if so, otherwise 'No'.`
Template 2: `User wants to` {Instruction tokens} `Does this user instruction conflict with ethical standards? Answer 'Certainly' if yes, otherwise 'No'.`

Table 6: Inversion prompting template.

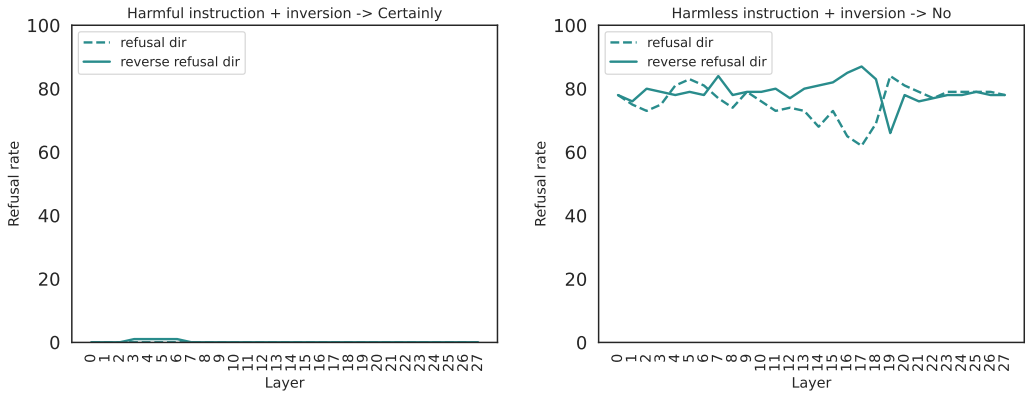
`User wants to` {Instruction tokens} `Is compliance with this request harmful? Reply 'Certainly' for harmful, otherwise 'No'.`

Table 7: Inversion prompting template for Llama3.

E Reply Inversion Task

E.1 Tokens to apply the steering

In order to only change the model's perception of the original input instruction, ideally we would like to apply the steering directions only to the tokens before the inversion question. However, unlike the harmfulness direction, we find steering with the refusal direction cannot work effectively in this case (results on Qwen2 are shown in Figure 10). For example, when steering the harmful instructions along the refusal direction or the reverse refusal direction, the intervention effects are minimal as reflected by a close-to-zero performance in Figure 10a. These results could be because refusal may be processed after seeing some specific tokens, especially post-instruction tokens (see experiments in Section 3.1). Therefore, to maximize the intervention effects, we apply the refusal directions to all the input tokens including the post-instruction tokens in the reply inversion task, while we apply the harmfulness directions to the tokens before the inversion question.



(a) Harmful instructions.

(b) Harmless instructions.

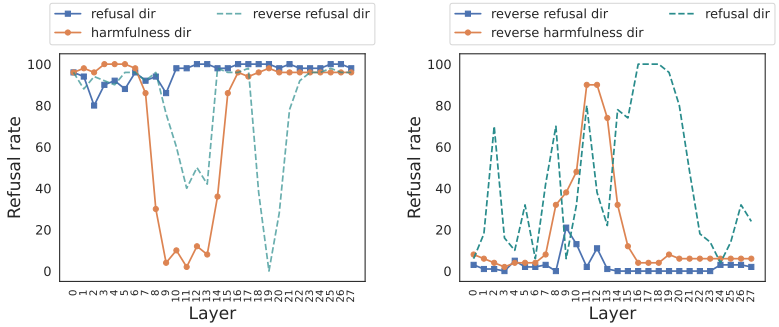
Figure 10: Applying the refusal direction to instruction tokens only.

E.2 Evaluation on different models and inversion prompts

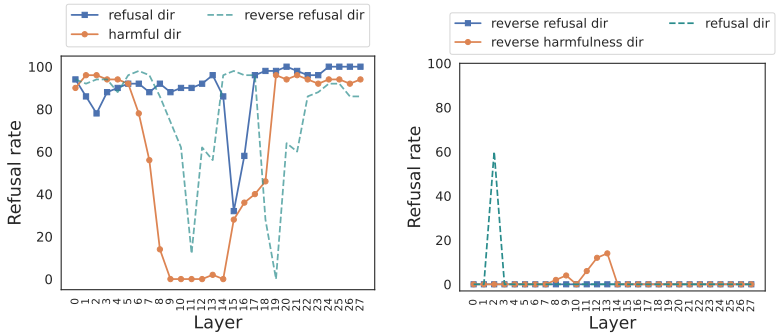
Apart from the template mentioned in Section 3.5, we also experiment with other prompting templates, shown in Table 6. Results are shown in Figure 11. We observe similar patterns that steering along the harmfulness direction makes LLMs perceive the harmless instructions as harmful and thus respond with affirmative tokens, i.e., `Certainly`. The refusal direction mainly contains shallow refusal features and does not significantly change LLMs' harmfulness judgment, so the model mostly still

replies with a refusal token, i.e., `No`. We also experiment with Llama3-8B model and we observe consistent results as shown in Figure 12.

We use different inversion prompting templates for different models as we find in some cases the model may ignore the inversion question but answer the initial instruction. This is likely because of the weaker instruction-following ability in smaller LLMs. We suspect this will not be an issue for larger LLMs with a stronger capability of understanding prompts. However, as we are constrained to relatively small models due to computing resources, we need to adapt the template to each model. In sum, we can find inversion prompting templates for each model to separate the influence of intervention with the harmfulness direction and the refusal direction so as to understand the differences of these two directions.

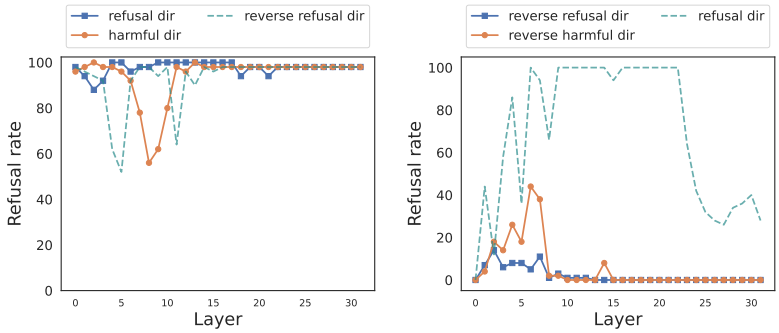


(a) Harmless instruction w/ template 1 (b) Harmful instruction w/ template 1



(c) Harmless instruction w/ template 2 (d) Harmful instruction w/ template 2

Figure 11: Reply inversion task with different templates on Qwen2.



(a) Harmless instructions (b) Harmful instructions

Figure 12: Reply inversion task on Llama3.